PART IIA, PAPER 3, SUPERVISION 5
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SOLUTIONS TO PROBLEMS

S5.1 Warning: kids are being used as an instrument for wages!
We are interested in estimating how women's hours of work respond to the wages they get. Consider the simultaneous equations relating wages earned and hours worked for women in the labour force:

$$\texttt{hours}_i = \beta_0 + \beta_1 \texttt{wage}_i + \beta_2 \texttt{kids}_i + u_i, \tag{1}$$
$$\texttt{wage}_i = \gamma_0 + \gamma_1 \texttt{hours}_i + \gamma_2 \texttt{kids}_i + v_i. \tag{2}$$

(a) Where do you think these equations come from, i.e. what are the economic rationale for these equations?

Solution: Expression (1) is the labour supply decision by workers, and the expression (2) is the labour demand decision of firms, who typically pay higher wages for those who work full time.

(b) Would an OLS of `hours` on `wage` and `kids` give us consistent estimates of the causal effect of wage on hours of work? Please justify your answer.

Solution: Not consistent due to simultaneity. Justification provided below.

(c) Solve for `hours` and `wage` in terms of `kids`, u and v by solving the equations (1) and (2). Now consider equation (1) and check if any of the Gauss-Markov assumptions are necessarily violated for this equation.

Solution: This is a classical example of simultaneous equations, with two variables being determined at the same time by two distinct behavioral process: the wife's decision to work and the husband's decision to work. Expressions in (1) and (2) are the structural equations, where the coefficients have theoretical / structural interpretation. These cannot be estimated using OLS due to endogeneity. In order to estimate the coefficients, we need to re-write these as the reduced-form equations. First, substitute for $\texttt{hours}_i$ in equation (2) and re-arrange

$$\begin{aligned}
\texttt{hours}_i &= \beta_0 + \beta_1 \left(\gamma_0 + \gamma_1 \texttt{hours}_i + \gamma_2 \texttt{kids}_i + v_i\right) + \beta_2 \texttt{kids}_i + u_i \\
(1 - \beta_1\gamma_1)\, \texttt{hours}_i &= \beta_0 + \beta_1\gamma_0 + \beta_1\gamma_2 \texttt{kids}_i + \beta_1 v_i + \beta_2 \texttt{kids}_i + u_i \\
\texttt{hours}_i &= \frac{\beta_0 + \beta_1\gamma_0}{1 - \beta_1\gamma_1} + \left[\frac{\beta_1\gamma_2 + \beta_2}{1 - \beta_1\gamma_1}\right] \texttt{kids}_i + \frac{\beta_1 v_i + u_i}{1 - \beta_1\gamma_1}.
\end{aligned} \tag{3}$$

Note the requirement of $\beta_1\gamma_1 \neq 0$ for the reduced form to exist. The regression that is actually being estimated does not of course include any of the structural parameters, but instead is specified as a reduced-form equation with reduced-form parameters:

$$\texttt{hours}_i = \pi_{10} + \pi_{11} \texttt{kids}_i + \varepsilon_{1i}, \tag{4}$$

The structural parameters $\beta_0, \beta_1, \gamma_0, \gamma_1$ and the unobserved structural shocks u_i, v_i, are related to the and the reduced-form coefficients π_{10}, π_{11} and residuals ε_{1i} as

$$\pi_{10} = \frac{\beta_0 + \beta_1\gamma_0}{1 - \beta_1\gamma_1}, \quad \pi_{11} = \frac{\beta_1\gamma_2 + \beta_2}{1 - \beta_1\gamma_1}, \quad \varepsilon_{1i} = \frac{\beta_1 v_i + u_i}{1 - \beta_1\gamma_1}.$$

Using the same method for $\texttt{wage}_i$ in equation (2):

$$\texttt{wage}_i = \frac{\gamma_0 + \gamma_1\beta_0}{1-\beta_1\gamma_1} + \left[\frac{\gamma_1\beta_2 + \gamma_2}{1-\beta_1\gamma_1}\right]\texttt{kids}_i + \frac{\gamma_1 u_i + v_i}{1-\beta_1\gamma_1}. \tag{5}$$

The reduced-from this is parameterized as

$$\texttt{wage}_i = \pi_{20} + \pi_{21}\texttt{kids}_i + \varepsilon_{2i}, \tag{6}$$

where

$$\pi_{20} = \frac{\gamma_0 + \gamma_1\beta_0}{1-\beta_1\gamma_1}, \quad \pi_{21} = \frac{\gamma_1\beta_2 + \gamma_2}{1-\beta_1\gamma_1}, \quad \varepsilon_{2i} = \frac{\gamma_1 u_i + v_i}{1-\beta_1\gamma_1}.$$

Now, let's get back to the original question and consider the OLS estimator of β_1 in equation (1). It will be biased for β_1, since `hours` is endogenous in regression (1). That follows from the reduced-form equation (5), which implies that $\text{Cov}\left(u_i, \texttt{wage}_i\right) \neq 0$. We can derive this explicitly and note that we assumed $\text{Cov}\left(u_i, \texttt{kids}\right) = 0$:

$$\begin{aligned}
\text{Cov}\left(u_i, \texttt{wage}_i\right) &= \text{Cov}\left(u_i, \frac{\gamma_0 + \gamma_1\beta_0}{1-\beta_1\gamma_1} + \left[\frac{\gamma_1\beta_2 + \gamma_2}{1-\beta_1\gamma_1}\right]\texttt{kids}_i + \frac{\gamma_1 u_i + v_i}{1-\beta_1\gamma_1}\right) \\
&= \text{Cov}\left(u_i, \left[\frac{\gamma_1}{1-\beta_1\gamma_1}\right]u_i + \left[\frac{1}{1-\beta_1\gamma_1}\right]v_i\right) \\
&= \left[\frac{\gamma_1}{1-\beta_1\gamma_1}\right]\text{Cov}\left(u_i, u_i\right) + \left[\frac{1}{1-\beta_1\gamma_1}\right]\text{Cov}\left(u_i, v_i\right) \\
&= \left[\frac{\gamma_1}{1-\beta_1\gamma_1}\right]\sigma_u^2
\end{aligned}$$

where $\text{Cov}\left(u_i, u_i\right) = \text{Var}\left(u_i\right) = \sigma^2 > 0$ and we assume $\text{Cov}\left(u_i, v_i\right) = 0$ for simplicity, as it is the assumption used in part (e). Clearly, in the original structural equation (1) wage is endogenous, since $\text{Cov}\left(u_i, \texttt{wage}_i\right) \neq 0$. This is a violation of the Gauss-Markov assumption of exogeneity of $\texttt{wage}_i$, required for the OLS estimators to be unbiased and consistent. The asymptotic bias / inconsistency of $\hat{\beta}_1$ will have the same sign as the coefficient $\gamma_1/\left(1-\beta_1\gamma_1\right)$. Similarly, equation (3) implies that $\text{Cov}\left(v_i, \texttt{hours}_i\right) \neq 0$, making `hours` endogenous in equation (2) and hence biasing the OLS estimator of γ_1.

(d) Under what conditions can we use `kids` as an instrument for `wage` and estimate β_1 by two-stage least squares? Can you write down one situation where these assumptions may fail?

Solution: The order condition must satisfied: we have one instrument for one endogenous variable. We want the rank condition to be satisfied, which will be the case when $\beta_2 = 0$ and $\gamma_2 \neq 0$, since this satisfies the exclusion restriction. This will imply that we can use `kids` as a valid instrument for the endogenous variable `wage` in equation (1) and hence identify (consistently estimate) β_1. However, we will no longer be able to use `kids` again as an instrument for another endogenous variable, and so we will need another IV in order to identify γ_1 in equation (2).

(e) Suppose that the covariance between u_i and v_i across households is zero, and suppose `kids` is a valid IV for `wage` in equation (1). Can you describe a way to consistently estimate γ_1, the coefficient of `hours` in the second equation, i.e., equation (2)?

Solution: This question is about the two-stage least squares procedure (2SLS), where we first use `kids` as the instrument for `wage` in equation (2) to consistently estimate β_1. In the first stage we estimate

$$\texttt{wage}_i = \pi_{10} + \pi_{11}\texttt{kids}_i + \varepsilon_{1i}, \quad i = 1, \ldots, n, \tag{7}$$

and save the predicted values of $\widehat{\texttt{wage}}_i$, which we then use in the second stage

$$\texttt{hours}_i = \beta_0 + \beta_1\widehat{\texttt{wage}}_i + u_i, \quad i = 1, \ldots, n, \tag{8}$$

to consistently estimate β_1. In order to identify γ_1 in equation (2), we could then use the predicted residuals from the above second-stage regression, namely

$$\hat{u}_i = \texttt{hours}_i - \hat{\beta}_0 + \hat{\beta}_1\texttt{wage}_i, \quad i = 1, \ldots, n. \tag{9}$$

These OLS fitted $\hat{u}_i$ will be consistent estimators of the structural unobserved shocks u_i. Also, it follows that these predicted residuals are correlated with $\texttt{hours}_i$, since

$$\text{Cov}\left(\hat{u}_i, \texttt{hours}_i\right) = \text{Cov}\left(\texttt{hours}_i - \hat{\beta}_0 + \hat{\beta}_1\texttt{wage}_i, \texttt{hours}_i\right) \neq 0.$$

Here, $\hat{u}_i$ are informative for $\texttt{hours}_i$ in equation (5). This can also be seen from the reduced-form equation (3). Moreover, since we are told that $\text{Cov}\left(u_i, v_i\right) = 0$, and know that $\hat{u}_i$ are consistent for u_i, $\hat{u}_i$ also satisfy the exclusion restriction for equation (2), so they are a valid instrument for $\texttt{hours}_i$. Therefore, $\hat{u}_i$ satisfy both conditions to be a good instrument for $\texttt{hours}_i$, which is endogenous in (2). We can then conduct a 2SLS estimation procedure, where in the first stage we estimate the following:

$$\texttt{hours}_i = \delta_0 + \delta_1\hat{u}_i + \varepsilon_i, \quad i = 1, \ldots, n, \tag{10}$$

and save the predicted values $\widehat{\texttt{hours}}_i = \hat{\delta}_0 + \hat{\delta}_1\hat{u}_i$. In the second stage, regress

$$\texttt{wage}_i = \gamma_0 + \gamma_1\widehat{\texttt{hours}}_i + v_i, \quad i = 1, \ldots, n \tag{11}$$

The OLS estimator $\hat{\gamma}_1$ from this regression will be consistent for γ_1.

S5.2 Defund the police: an empirical investigation.
We are interested in the determinants of crime.

(a) Download the dataset `crime4.dta`. See what the variables mean by typing `des`. Summarize the data using the command `summ`.

(b) Run an OLS regression using `reg crmrte polpc west central urban`. What do you infer from the row corresponding to `west`. Interpret the positive and highly significant coefficient on `polpc`.

Solution: The following `STATA` code is used to estimate the regression and the estimated coefficients together with the corresponding standard errors are presented in column (1) of Table 1.

```
reg crmrte polpc west central urban
```

We might expect that hiring more police officers should help fight crime, and therefore the coefficient on police in the underlying regression should be negative. However, the OLS coefficient on `polpc` is positive and significant, suggesting that regions with higher numbers of police officers have higher crime rates. This is counter-intuitive, since if the interpretation is causal and runs from police to crime, this results would suggests that hiring more police officers increases the crime rate.

Table 1: OLS and IV estimation of `crmrte` on `polpc`

	(1) OLS	(2) IV
`polpc`	1.291	2.315
	(0.1961)	(1.7095)
`west`	−0.015	−0.015
	(0.0014)	(0.0015)
`central`	−0.002	−0.002
	(0.0012)	(0.0013)
`urban`	0.034	0.034
	(0.0019)	(0.0019)
`const`	0.0304	0.029
	(0.0009)	(0.0032)
F-stat	130.02	2.71
R^2	0.4542	0.0170

standard errors in parenthesis

(c) Explain why `polpc` is likely endogenous in your regression. We intend to use `taxpc` as an IV for `polpc`. In terms of a simultaneous equations model, explain which coefficients should be zero or non-zero, in order for `taxpc` to be a valid IV for `polpc`. Be careful about what equation is being identified.

Solution: The positive coefficient on police per capita in the previous regression contradict economic theory (and common sense). This may be the case if the police themselves are involved in crime. However, a it is more likely that the estimator is biased. The reason for the bias is the endogeneity of `polpc` with respect to the unobserved error terms in the regression. In turn, endogeneity could be present because causality in this relationship runs in the opposite direction - increasing crime rate leads to more hiring into the police force. Overall, there is likely to be simultaneity in the relationship between police and crime rates. As we know, bi-directional causality causes endogeneity and is a major source of bias in the OLS estimator. In our case we suspect that the bias is positive and large in magnitude.

In order to correct for the estimation bias we use an instrument `taxpc` for police per capita and run a 2-stage least squares estimation. The following system of simultaneous equations will be estimated:

$$\texttt{polpc}_i = \gamma_0 + \gamma_1\texttt{crmrte}_i + \gamma_2\texttt{taxpc}_i + \delta_{11}\texttt{west}_i + \delta_{12}\texttt{central}_i + \delta_{13}\texttt{urban}_i + v_i$$
$$\texttt{crmrte}_i = \beta_0 + \beta_1\texttt{polpc}_i + \beta_2\texttt{taxpc}_i + \delta_{11}\texttt{west}_i + \delta_{12}\texttt{central}_i + \delta_{13}\texttt{urban}_i + u_i$$

For identification of β_1 in the second regression we need (1) $\gamma_2 \neq 0$, which means that tax rate is informative for police per capita (order condition); and (2) $\beta_2 = 0$, which is the exclusion restriction implying that tax rate is exogenous in the second regression (rank condition).

(d) Run a 2SLS using `taxpc` as an IV. Compare with the OLS output.

Solution: The following `STATA` code is used to estimate the IV (2SLS) regression and the estimated coefficients together with the corresponding standard errors are presented in column (2) of Table 1.

```
ivregress 2sls crmrte west central urban (polpc = taxpc), first
```

The coefficient on `polpc` is still positive, larger in magnitude, but insignificant. Also notice a large drop in R^2 and overall regression significance, as suggested by the F-statistic. This is to be expected, since by instrumenting `polpc` with `taxpc` in the second-stage regression (where $\widehat{\texttt{polpc}}$ from the first stage is used instead of the observed `polpc`) we are only keeping the exogenous variation in police per capita. This is considerably lower than the overall variation in `polpc`, and so the overall explanatory power of the regression falls. Since this reduction is so dramatic, we suspect that the instrument was chosen poorly and does not help us solve the endogeneity problem.

(e) If `taxpc` were a valid IV, what does the 2SLS estimate say about the effect of police on crime? What does it say about the effect of crime on police employment? Can you suggest a reason why `taxpc` may not be a valid instrument? If `taxpc` is indeed a valid IV, how would you test the endogeneity of `polpc` in the OLS regression in (d). Perform this test and write down your conclusion.

Solution: Taxes may be raised in places with high crime rates to employ more police, violating the exclusion restriction, that is $\beta_2 \neq 0$. To test for endogeneity of `polpc` in the second-stage regression use he Wu-Hausman test, which hypothesizes no endogeneity of `polpc`, or more formally that the OLS estimator is equivalent to the IV estimator. We use the following `STATA` code to run the test using the endogeneity test command and also manually (which of course yield the same result):

```
// Wu-Hausman and Durbin test for endogeneity of polpc
estat endogenous

// Test endogeneity of polpc manually
reg polpc taxpc west central urban
predict res, res
reg crmrte polpc west central urban res
test res
```

The F-test statistic for this test has a p-value of 0.5398 and we do not reject the null of no endogeneity: OLS and IV estimators are statistically equivalent (at least if tax is used as the instrument). We conclude that both estimators are either valid or invalid.

S5.1 Treating the untreated out of poverty.

The file `nsw.dta` contains data on a sample of 722 men who participated in the Na-

tional Supported Work (NSW) experiment. The NSW was a temporary employment program designed to help disadvantaged workers lacking basic job skills move into the labour market by giving them work experience and counseling in a sheltered environment. The NSW program assigned qualified applicants to training positions randomly. Those assigned to the treatment group (297 men) received all the benefits of the NSW program. Those assigned to the control group (425 men) were left to fend for themselves. For those assigned to the treatment group, the program guaranteed a job for 9 to 18 months. The study collected demographic data and earnings from the treatment and the control group members at the baseline, `incbefore`, and after the end of the experimental period, `incafter`. We would like to understand whether the NSW employment program had a positive (causal) effect on the earnings of the participants.

First, suppose we did not have a randomized controlled trial available, but instead had observational data for a sample of individuals some of whom had opted for training and some had not.

(a) In the context of the evaluation of treatment effects, what is selection bias?

Solution: Let $D_i \in \{0, 1\}$ denote the treatment variable (called `treated` in the dataset) equal to 1 if i has participated in the NSW (treatment group) and 0 otherwise (control group). For each i denote potential outcomes by $Y_i(0)$ and $Y_i(1)$: $Y_i(0)$ is the income of person i if they do not receive the treatment, and $Y_i(1)$ if they do, where Y_i is the observed variable `incafter`. Then $Y_i(1) - Y_i(0)$ is the causal effect of the treatment for individual i. The average treatment effect is given by

$$\text{ATE} = \mathbb{E}\left[Y_i(1)\right] - \mathbb{E}\left[Y_i(0)\right].$$

The average effect of treatment on the treated (ATT) is give by

$$\text{ATT} = \mathbb{E}\left[Y_i(1) \mid D_i = 1\right] - \mathbb{E}\left[Y_i(0) \mid D_i = 1\right].$$

The observational difference, the quantity that can be measured, is

$$\mathbb{E}\left[Y_i(1) \mid D_i = 1\right] - \mathbb{E}\left[Y_i(0) \mid D_i = 0\right]$$

It can also be expressed as

$$\left\{\mathbb{E}\left[Y_i(1) \mid D_i = 1\right] - \mathbb{E}\left[Y_i(0) \mid D_i = 1\right]\right\} + \left\{\mathbb{E}\left[Y_i(0) \mid D_i = 1\right] - \mathbb{E}\left[Y_i(0) \mid D_i = 0\right]\right\}.$$

The term in the first curly bracket is the ATT and the term in the second bracket is the selection bias. The selection bias is the difference between the income after the program for the untreated individuals had they participated in the NSW (unobserved countefactual) and their actual `incafter` (observed).

(b) Give an intuitive argument for the sign of the selection bias on the effects of training programs like NSW on earnings.

Solution: Imagine the assignment to the NSW program was not random, and instead individuals made a personal choice to participate. Then we can make an argument that those who chose to participate are more motivated to receive training and job experience to improve their earnings than those who decided to opt out. As motivation is positively correlated with baseline earnings, this would imply that $\mathbb{E}\left[Y_i(0) \mid D_i = 1\right] > \mathbb{E}\left[Y_i(0) \mid D_i = 0\right]$ and the selection bias in this case is positive. The observational difference would then overestimate the causal effect.

(c) Suggest an instrument which might help you identify the causal effect of interest.

Solution: To estimate the treatment effect we could estimate a regression of `incafter` on the treatment assignment variable `treated` (D_i) and the control variables $\mathbf{X}$, which include age, education, ethnic background, and income before the program:

$$Y_i = \beta_0 + \delta D_i + \mathbf{X}^\intercal \boldsymbol{\beta} + u_i$$

The effect of the NSW program would then be visible in δ. However, since we assumed $\text{Cov}(D_i, u_i) > 0$, as u_i contains unobserved factors such as motivation, and since we expect $\delta > 0$, the OLS estimator of δ will be biased upwards. We are then looking for an instrument Z_i for endogenous variable D_i, which would be (as good as) randomly assigned, and have $\text{Cov}(Z_i, D_i) \neq 0$ and $\text{Cov}(Z_i, u_i) = 0$. Note that for the IV to be good, Z_i must only affect ΔY_i through its effect on D_i, that is we cannot have Z_i influence ΔY_i directly and cannot have confounding factors that affect both Z_i and ΔY_i.

Now, let's get back to the NSW experiment.

(d) Provide empirical support for the claim that participants were assigned randomly to the treatment and control groups.

Solution: For random assignment we would expect the mean values of the variables (excluding incomes after the program) in the treatment and the control groups to be the same. So we can conduct two-sample tests for the equality of means of the variables in the treated and the untreated groups. For example, for the age variable test

$$H_0 : \mathbb{E}[\texttt{age}_i \mid D_i = 1] = [\texttt{age}_i \mid D_i = 0]$$
$$H_1 : \mathbb{E}[\texttt{age}_i \mid D_i = 1] \neq [\texttt{age}_i \mid D_i = 0]$$

using sample means $\overline{\texttt{age}}_\text{t} = \overline{\texttt{age}}_\text{c}$ and corresponding standard errors, where $\overline{\texttt{age}}_\text{t}$ and $\overline{\texttt{age}}_\text{c}$ are the sample mean ages in the treatment and control groups, respectively. The results for the two-sided tests are presented in the table below, all of which do not reject the null at 5%, so that we have sufficient evidence of random assignment. Note that since `hisp` and `married` are dummy variables, we test for equality of proportions. Clearly, we do not need to conduct the test for the outcome variable, `incafter`.

Variable	t	p-value	Conclusion at 5%
`age`	-0.3565	0.7216	Do not reject
`educ`	-1.4620	0.1443	Do not reject
`black`	-0.0446	0.9645	Do not reject
`hisp`	0.8042	0.4213	Do not reject
`married`	-0.3840	0.7009	Do not reject
`incbefore`	-0.1040	0.9172	Do not reject

(e) Calculate the following quantities:

i. the treatment-assignment probabilities and their standard errors;
Solution: $\mathbb{P}(D_i = 1) = 0.411$ and $\mathbb{P}(D_i = 0) = 0.589$ with standard errors 0.118.

ii. the average outcome (post program earnings) and its standard error for the treated;
Solution: $\mathbb{E}[Y_i(1) \mid D_i = 1] = 5976.352$ with standard error of 401.75, since there are 297 observations in the treatment group.

iii. the average outcome (post program earnings) and its standard error for the untreated;
Solution: $\mathbb{E}[Y_i(0) \mid D_i = 0] = 5090.048$ with standard deviation of 277.37, since there are 425 observations in the control group.

iv. the average treatment effect/average effect of treatment on the treated and its 95% confidence interval;
Solution: As the randomized controlled trial ensures there is no selection bias,

$$\text{ATT} = \text{ATE} = \text{Obs. Diff.} = \mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] = 886.31$$

The standard error is 488.20 and the 95% confidence interval is $[-70.57, 1843.18]$ which includes zero. However the 90% confidence interval $[83.21, 1689.40]$ does not.

(f) Test the hypothesis that the treatment had a positive effect.
Solution: We are testing if the ATT is greater than zero, a one-sided test with the test statistic under the null of no significance impact of the treatment and an alternative hypothesis of a positive effect:

$$t = \frac{886.31}{488.20} = 1.815$$

Since the sample is large, we can use standard normal approximation critical value of $z_{0.95} = 1.645$ and reject the null at 5% significance (evidence of a positive effect of treatment), but reject it at 1% with a critical value of $z_{0.99} = 2.326$ (no positive effect of treatment). We could also regress $Y_i =$ `incafter` on the treatment variable and controls:

$$Y_i = \beta_0 + \delta D_i + \mathbf{X}^\intercal \boldsymbol{\beta} + u_i$$

and estimate the treatment effect δ using OLS. The coefficient on `treated` is 830.67, so pretty close to the unconditional treatment effect in (b), and it has a p-value of 0.075. The conclusion about the positive effect of the NSW program is again marginal: significant at 5%, but insignificant at 1% (for a one-sided test). Due to random assignment to treatment, which seems to be validated by our results in (a), we can be somewhat certain that these estimates do not contain an endogeneity bias.

(g) Is there any evidence of heterogeneous treatment effects for specific subgroups of the population?
Solution: Compute treatment effects for different subgroups of the population and test the null hypothesis that they are statistically greater than zero:

Group	ATT	Std. error	t	Conclusion at 5%
Married	2333.148	1234.204	1.8904	Reject
Black	1159.423	559.9809	2.0705	Reject
Hispanic	-429.406	1326.238	-0.3238	Do not reject
No degree	706.465	539.741	1.309	Do not reject

(h) What do you conclude from the experiment? Critically evaluate whether your results imply that employment programs like the NSW can be used more widely to improve the outcomes of disadvantaged workers.

Solution: There is evidence that assignment to the treatment group was random. We have borderline statistical evidence that the program has been successful (only at 5%, but not at 1%). There is also evidence of heterogeneity of the treatment in different sub-groups of the population. The program seems to be more effective for those individuals who are married and/or black, while for individuals who are Hispanic or have no degree it is not so effective.

S5.4 "As you estimate the OLS and say, take my mean away."

A variable Y_i depends on a non-stochastic variable X_i with the relationship

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \ldots, n, \tag{12}$$

where u_i are the error terms. Given a random sample of n observations on X_i and Y_i, a researcher decides to estimate β_1 using the following estimator

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2}. \tag{13}$$

(a) Demonstrate that $\tilde{\beta}_1$ is in general a biased estimator of β_1 and derive this bias. Explicitly state any assumptions that you require in your derivation.

Solution: The proposed estimator $\tilde{\beta}_1$ does not de-mean the variables, and so will generally be biased, unless X_i has zero mean. In order to show that $\tilde{\beta}_1$ is biased, substitute (12) into (13)

$$\begin{aligned}
\tilde{\beta}_1 &= \frac{\sum_{i=1}^n X_i Y_i}{\sum_{i=1}^n X_i^2} = \frac{\sum_{i=1}^n X_i \left(\beta_0 + \beta_1 X_i + u_i\right)}{\sum_{i=1}^n X_i^2} \\
&= \beta_0 \frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i^2} + \beta_1 \frac{\sum_{i=1}^n X_i^2}{\sum_{i=1}^n X_i^2} + \frac{\sum_{i=1}^n X_i u_i}{\sum_{i=1}^n X_i^2} \\
&= \beta_1 + \beta_0 \frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i^2} + \frac{\sum_{i=1}^n X_i u_i}{\sum_{i=1}^n X_i^2}
\end{aligned}$$

Take the expectation of this expression:

$$\begin{aligned}
\mathbb{E}\left[\tilde{\beta}_1\right] &= \mathbb{E}\left[\beta_1 + \beta_0 \frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i^2} + \frac{\sum_{i=1}^n X_i u_i}{\sum_{i=1}^n X_i^2}\right] \\
&= \beta_1 + \mathbb{E}\left[\beta_0 \frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i^2}\right] + \mathbb{E}\left[\frac{\sum_{i=1}^n X_i u_i}{\sum_{i=1}^n X_i^2}\right] \\
&= \beta_1 + \beta_0 \left[\frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i^2}\right]
\end{aligned}$$

where we used that X_i is nonstochastic, i.e. fixed / non-random, and assumed that $\mathbb{E}\left[X_i u_i\right] = 0$, that is that X_i is exogenous. Hence, the bias will be equal to

$$\text{Bias}\left(\tilde{\beta}_1\right) = \mathbb{E}\left[\tilde{\beta}_1\right] - \beta_1 = \beta_0 \left[\frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i^2}\right]. \tag{14}$$

(b) Discuss whether it is possible to determine the sign of the bias in $\tilde{\beta}_1$. Under what conditions will $\tilde{\beta}_1$ be unbiased.

Solution: The bias will depend on the sign of β_0 and the population mean of the explanatory variable. The denominator is the estimate of the sample variance, so will always be positive. The bias in $\tilde{\beta}_1$ will then be positive if the signs of β_0 and $\mathbb{E}[X_i]$ are the same, and negative when the signs are opposite. The bias will be zero if $\beta_0 = 0$, and there is no intercept in the model, that is $\mathbb{E}[Y_i|X_i = 0] = 0$, or if $\sum_{i=1}^n X_i = 0$, which can only happen if $\mathbb{E}[X_i] = 0$.

(c) Suggest an unbiased linear estimator of β_1 that has the smallest variance out all possible unbiased linear estimators. Under what conditions does that property hold?

Solution: Under the assumptions of the Gauss-Markov theorem, the OLS estimator of β_1 will be unbiased:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n \left(X_i - \bar{X}\right) Y_i}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} = \sum_{i=1}^n \omega_i^{\text{OLS}} Y_i, \qquad \omega_i^{\text{OLS}} = \frac{\left(X_i - \bar{X}\right)}{\sum_{i=1}^n \left(X_i - \bar{X}\right)^2} \tag{15}$$

Note that the estimator from (13) is also linear, since it is expressed as

$$\tilde{\beta}_1 = \sum_{i=1}^n \tilde{\omega}_i Y_i, \qquad \tilde{\omega}_i = \frac{X_i}{\sum_{i=1}^n X_i^2}. \tag{16}$$

The Gauss-Markov theorem also says that $\text{Var}(\hat{\beta}_1) \leq \text{Var}(\beta_1^*)$, where β_1^* is any linear unbiased estimator

$$\beta_1^* = \sum_{i=1}^n \omega_i Y_i. \tag{17}$$

The assumptions that we required are

(A1) Linearity: the population model is linear in parameters, as in (12).
(A2) Fixed or random regressors: X_i, for $i = 1, \ldots, n$, with $\text{Var}(X_i) > 0$.
(A3) Exogeneity: $\mathbb{E}[u_i|X_i] = 0$, for $i = 1, \ldots, n$.
(A4) Homoskedasticity: $\text{Var}(u_i|X_i) = \sigma^2$, for $i = 1, \ldots, n$.
(A5) No serial correlation: $\mathbb{E}[u_i u_j|X_i, X_j] = 0$, for $i \neq j$.

S5.5 Cobb & Douglas's excellent adventure continues (this time by plane)

This question asks you to look at the production technology in US air transport over the period 1948-79. The `airtran.dta` dataset contains the following variables that are all available for the full sample period 1948-79:

`y`	output measure for air transport
`r`	cost of capital in air transport
`k`	capital stock employed in air transport
`w`	wage rate index for air transport
`l`	labour employed in air transport
`c`	intercept
`time`	trend term (time t)

Output results in Table 2 were generated using the following `STATA` code:

```
>> eststo clear
>> eststo:  quietly reg ly lk ll
>> eststo:  quietly reg lyl lkl ll
>> eststo:  quietly reg lkl lwr ly
>> eststo:  quietly reg depvar lwr ly
>> eststo:  quietly reg lkl lwr ly if year <= 1963
>> eststo:  quietly reg lkl lwr ly if year > 1963
```

Table 2: `STATA` output for Cobb-Douglas regressions

	(1) `ly`	(2) `lyl`	(3) `lkl`	(4) `lklwr`	(5) `lkl`	(6) `lkl`
`lk`	0.0108					
	(0.16)					
`ll`	1.771***	0.781***				
	(12.84)	(10.32)				
`lkl`		0.0108				
		(0.16)				
`lwr`			0.564***	−0.436***	0.428*	0.660***
			(7.49)	(−5.79)	(2.23)	(8.59)
`ly`			0.248***	0.248***	0.118	−0.167
			(4.69)	(4.69)	(1.48)	(−1.18)
`const`	−0.192	−0.192	−1.175***	−1.175***	−1.053***	−0.0870
	(−1.65)	(−1.65)	(−14.48)	(−14.48)	(−10.35)	(−0.26)
Sample	full	full	full	white	$t \leq 1963$	$t > 1963$
N	32	32	32	32	16	16
R^2	0.989	0.946	0.925	0.536	0.417	0.935
adj. R^2	0.988	0.943	0.920	0.504	0.327	0.925
F-stat	1310.4	256.1	179.3	16.76	4.643	93.08
SSR	0.282	0.282	0.778	0.778	0.368	0.121
RMSE	0.0986	0.0986	0.164	0.164	0.168	0.0966

t statistics in parentheses

$^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

(a) Compute the natural logarithms of the variables in the dataset and name them as $\texttt{ly}_i = \ln(\texttt{y}_i)$, $\texttt{lr}_i = \ln(\texttt{r}_i)$ and so on. Use your computed logarithmic data to estimate the Cobb-Douglas production function (log-linear model)

$$\texttt{ly}_i = \beta_0 + \beta_1 \texttt{lk}_i + \beta_2 \texttt{ll}_i + u_i. \tag{18}$$

Solution: Output in column (1) of Table 2 is generated using the following code:

```
>> gen ly = ln(y)
>> gen lk = ln(k)
>> gen ll = ln(l)
>> reg ly lk ll
```

(b) Reparametrize your model in a way which allows you to test the hypothesis that $\beta_1 + \beta_2 = 1$. Estimate the transformed model and test this hypothesis, using a 5%-significance level. Interpret your results with reference to economic theory.

Solution: We want to test $H_0\colon \beta_1 + \beta_2 = 1$ (constant returns to scale) against $H_0\colon \beta_1 + \beta_2 \neq 1$ in regression 2, which can be done directly in `STATA` using the code below and with the F-statistic of 106.45 and a rejection of the null at all imaginable significance levels.

```
>> reg ly lk ll
>> test lk + ll = 1
```

Another way of testing involves reparametrization of the model. Consider $\gamma = \beta_1 + \beta_2 - 1$, so that $\beta_2 = \gamma - \beta_1 + 1$, and the specification in (18) becomes:

$$\begin{aligned}\mathtt{ly}_i &= \beta_0 + \beta_1 \mathtt{lk}_i + \beta_2 \mathtt{ll}_i + u_i \\ &= \beta_0 + \beta_1 \mathtt{lk}_i + (\gamma - \beta_1 + 1)\,\mathtt{ll}_i + u_i \\ &= \beta_0 + \beta_1 (\mathtt{lk}_i - \mathtt{ll}_i) + (\gamma + 1)\,\mathtt{ll}_i + u_i\end{aligned}$$

Now we can move $\mathtt{ll}_i$ from the right to the left-hand side to obtain the re-paramtization of the following form:

$$\begin{aligned}(\mathtt{ly}_i - \mathtt{ll}_i) &= \beta_0 + \beta_1 (\mathtt{lk}_i - \mathtt{ll}_i) + \gamma \mathtt{ll}_i + u_i \\ \mathtt{lyl}_i &= \beta_0 + \beta_1 \mathtt{lkl}_i + \gamma \mathtt{ll}_i + u_i \qquad (19)\end{aligned}$$

In this regression, $\mathtt{lyl}_i = (\mathtt{ly}_i - \mathtt{ll}_i)$ is the new dependent variable, while $\mathtt{lkl}_i = (\mathtt{lk}_i - \mathtt{ll}_i)$ and $\mathtt{ll}_i$ are the two explanatory variables. Then, the test of $\beta_1 + \beta_2 = 0$ in (18) is equivalent to testing $H_0\colon \gamma = 0$ against $H_1\colon \gamma \neq 0$ in (19). Use the following code and report the output of the regression in column (2) of Table 2.

```
>> gen lyl = ly - ll
>> gen lkl = lk - ll
>> reg lyl lkl ll
```

From `STATA` output, $t_{\hat{\gamma}} = 10.32$. For $\alpha = 0.05$, the critical value for t_{29} is 2.045. Therefore, we reject the null hypothesis that $\beta_2 + \beta_3 = 1$ at 5% significance. Alternatively we can also define $\delta = \beta_1 + \beta_2$, in this case $\beta_1 = \delta - \beta_2$ and model 18 becomes

$$\mathtt{ly}_i = \beta_0 + \delta \mathtt{lk}_i + \beta_3 (\mathtt{ll}_i - \mathtt{lk}_i)\,. \qquad (20)$$

We use a t-test on δ with null hypothesis $H_0\colon \delta = 1$ against $H_1\colon \delta \neq 1$.

(c) Define the variables $\mathtt{lkl}_i = (\mathtt{lk}_i - \mathtt{ll}_i)$ and $\mathtt{lwr}_i = (\mathtt{lw}_i - \mathtt{lr}_i)$. Now estimate the following linear regression

$$\mathtt{lkl}_i = \gamma_0 + \gamma_1 \mathtt{lwr}_i + \gamma_2 \mathtt{ly}_i + v_i. \qquad (21)$$

What is the economic interpretation of this regression? [Hint: assume that the production function is Cobb-Douglas and work out the first-order conditions for producing `y` at minimum cost]. What are the hypotheses that this interpretation suggests should be tested? Re-parameterize the equation so that the relevant hypotheses can be tested, and then test them and report your results.

Solution: This regression shows whether, for the same level of output, an increase in the price of labour relative to the price of capital leads to employment of more capital-intensive techniques. In `STATA` use the following code and report the results in column (3) of Table 2:

```
>> gen lwr = ln(w) - ln(r)
>> reg lkl lwr ly
```

The coefficient on $\texttt{lwr} = \texttt{lw}_i - \texttt{lr}_i$ is positive and significant, meaning that an increase in $\texttt{lwr}_i$ shifts the production toward more capital intensive techniques. Equation (21) is the difference of log of conditional factor demand functions. Each factor demand function (K, L) is a function of factors price (w, r) and output (Y). The conditional factor demand function is derived by minimizing cost of production by choosing K and L according to the following minimization problem:

$$\begin{aligned} &\min_{K,L} wL + rK \\ &\text{s.t. } Y - F(K, L) = 0 \end{aligned}$$

The Lagrangian associated with this problem can be written as

$$\mathcal{L} = wL + rK + \lambda \left(Y - F(K, L)\right)$$

The first order conditions:

$$\frac{\partial \mathcal{L}}{\partial K} = r - \lambda F_K = 0 \tag{22}$$

$$\frac{\partial \mathcal{L}}{\partial L} = w - \lambda F_L = 0 \tag{23}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = Y - F(K, L) = 0 \tag{24}$$

where $F_K = \frac{\partial F}{\partial K}$ and $F_L = \frac{\partial F}{\partial L}$. From (22) and (23) we have

$$\frac{F_L}{F_K} = \frac{w}{r}. \tag{25}$$

In case of the Cobb-Douglas production function $Y = AK^\alpha L^\beta$, or alternatively the log transformation of the production function $\ln Y = \beta_0 + \beta_1 \ln K + \ln L$ where $A = \beta_0, \alpha = \beta_1$ and $\beta = \beta_2$. we will have

$$\frac{w}{r} = \frac{A\beta K^\alpha L^{\beta-1}}{A\alpha K^{\alpha-1} L^\beta} = \frac{\beta}{\alpha}\frac{K}{L}. \tag{26}$$

In logs this looks like this:

$$\ln w - \ln r = \log\left(\frac{\beta}{\alpha}\right) + \ln K - \ln L.$$

or equivalently

$$\ln K - \ln L = -\log\left(\frac{\beta}{\alpha}\right) + \ln w - \ln r. \tag{27}$$

Hence, at the equilibrium (27), the elasticity of substitution is equal to 1, and the conditional factor demands are independent of output. Comparing the equilibrium (27) with the empirical regression model (21), we might want to test whether air transport produces at minimum cost, (i.e. equilibrium) which means that $\gamma_1 = 1$ and $\gamma_2 = 0$ in regression model (21). So we want to test

$$\begin{aligned} H_0\colon\ & (\gamma_1 = 1) \cap (\gamma_2 = 0) \\ H_1\colon\ & (\gamma_1 \neq 1) \cup (\gamma_2 \neq 0) \end{aligned}$$

To do that, we can re-parameterize the model so that $\theta = \gamma_1 - 1$. Then $\gamma_1 = \theta + 1$, and model (21) becomes

$$\begin{aligned} \texttt{lkl}_i &= \gamma_0 + \gamma_1 \texttt{lwr}_i + \gamma_2 \texttt{ly}_i + v_i \\ \texttt{lkl}_i &= \gamma_0 + (\theta + 1)\,\texttt{lwr}_i + \gamma_2 \texttt{ly}_i + v_i \\ (\texttt{lkl}_i - \texttt{lwr}_i) &= \gamma_0 + \theta \texttt{lwr}_i + \gamma_2 \texttt{ly}_i + v_i \end{aligned} \tag{28}$$

Therefore, the hypothesis in the transformed equation (28) is

$$\begin{aligned} H_0\colon\ & (\theta = 0) \cap (\gamma_2 = 0) \\ H_1\colon\ & (\theta \neq 0) \cup (\gamma_2 \neq 0) \end{aligned}$$

which can be done using an F-test. Use the following `STATA` code and report the output of the regression in column (4) of Table 2:

```
>> gen lklwr = lkl - lwr
>> reg lklwr lwr ly
```

The F-statistic for the joint significance of the regressors is 16.76. Therefore we reject the null hypothesis and conclude that the production is not at the minimal cost.

(d) Re-estimate the unrestricted equation from (c), splitting the sample into the two sub-samples: 1948-63 and 1964-79. Test the hypothesis that the coefficients of the unrestricted equation are the same for both sub-periods.

Solution: We estimate the model in (21) three times:

$$\begin{aligned} \texttt{lkl}_i &= \gamma_0 + \gamma_1 \texttt{lwr}_i + \gamma_2 \texttt{ly}_i + v_i,\ i = 1, \ldots, n,\ SSR \\ \texttt{lkl}_i &= \gamma_0^1 + \gamma_1^1 \texttt{lwr}_i + \gamma_2^1 \texttt{ly}_i + v_i^1,\ i = 1, \ldots, n_1,\ SSR_1 \\ \texttt{lkl}_i &= \gamma_0^2 + \gamma_1^2 \texttt{lwr}_i + \gamma_2^2 \texttt{ly}_i + v_i^2,\ i = n_1 + 1, \ldots, n,\ SSR_2 \end{aligned}$$

We are testing stability of the coefficients between two periods, namely:

$$\begin{aligned} H_0\colon\ & \left(\gamma_0^1 = \gamma_0^2\right) \cap \left(\gamma_1^1 = \gamma_1^2\right) \cap \left(\gamma_2^1 = \gamma_2^2\right) \\ H_1\colon\ & \left(\gamma_0^1 \neq \gamma_0^2\right) \cup \left(\gamma_1^1 \neq \gamma_1^2\right) \cup \left(\gamma_2^1 = \gamma_2^2\right) \end{aligned}$$

Use the Chow test with $k = 2$ and $n = 32$ and the following test statistic

$$V = \frac{\left[SSR - (SSR_1 + SSR_2)\right] / (k+1)}{(SSR_1 + SSR_2) / (n - 2(k+1))} \sim F_{k+1, n-2(k-1)}.$$

Reject the null at α if the observed test statistic $v > F_{3,26} = 2.98$. Use the following `STATA` code to estimate the following regressions and record the respective SSR, the results are reported of Table 2: columns (3) for the full sample, column (5) for the earlier sub-sample and column (6) for the later sub-sample:

```
>> gen year = 1947 + time
>> reg lkl lwr ly
>> reg lkl lwr ly if time <= 1963
>> reg lkl lwr ly if time > 1963
```

The test statistic is then given by

$$v = \frac{[0.778 - (0.368 + 0.121)]/3}{(0.368 + 0.121)/26} = 5.124.$$

This is greater than the critical value and the null is rejected at 5%.

S5.6 Netflix and Amazon Prime are slowly making this question obsolete.
A researcher has the following data for 40 cities in the United Kingdom for the year 2002: T_i, annual total sales of cinema tickets per household, and P_i, the average price of a cinema ticket in the city, and Y_i, average household income. She believes that the true relationship is

$$\log T_i = \beta_1 + \beta_2 \log P_i + \beta_3 \log Y_i + u_i. \tag{29}$$

Given that data are not available for Y_i she fits the regression (estimated standard errors in parentheses):

$$\log T_i = \underset{(0.52)}{13.74} + \underset{(0.23)}{0.17} \log P_i, \quad R^2 = 0.01. \tag{30}$$

If the researcher had been able to obtain data on Y_i, the estimated regression equation would have been

$$\log T_i = -\underset{(2.93)}{1.63} - \underset{(0.21)}{0.48} \log P_i + \underset{(0.25)}{1.83} \log Y_i, \quad R^2 = 0.44. \tag{31}$$

Note: Y_i and P_i are positively correlated.

(a) Explain analytically whether $\hat{\beta}_2$ in (30) is likely to be biased.
Solution: The true model is given in (29), while the estimated regression with the omitted variable $\log Y_i$ can be written in the following theoretical form:

$$\log T_i = \delta_1 + \delta_2 \log P_i + v_i, \tag{32}$$

where v_i is an error term with zero mean and constant variance. The OLS estimator of δ_1 in regression (32) is given by the standard formula, where we then substitute for $\log T_i$ from the DGP (29):

$$\hat{\delta}_2 = \frac{\widehat{\text{Cov}}(\log P_i, \log T_i)}{\widehat{\text{Var}}(\log P_i)} = \frac{\widehat{\text{Cov}}(\log P_i, \beta_1 + \beta_2 \log P_i + \beta_3 \log Y_i + u_i)}{\widehat{\text{Var}}(\log P_i)}$$

$$\hat{\delta}_2 = \beta_2 + \beta_3 \left[\frac{\widehat{\text{Cov}}\left(\log P_i, \log Y_i\right)}{\widehat{\text{Var}}\left(\log P_i\right)} \right] + \frac{\widehat{\text{Cov}}\left(\log P_i, u_i\right)}{\widehat{\text{Var}}\left(\log P_i\right)}. \tag{33}$$

Clearly, $\hat{\delta}_2 \neq \beta_2$ and the OLS estimator in (30) is biased, unless the additional terms in the above expression are zero. If $\log P_i$ is exogenous in the original regression, then $\text{Cov}\left(\log P_i, u_i\right) = 0$, but in small samples, the estimated covariance $\widehat{\text{Cov}}\left(\log P_i, \log Y_i\right)$ is still likely to be non-zero. Thus, the third term should be zero in expectation. The second term in (33) is zero if either $\log P_i$ and $\log Y_i$ are uncorrelated, which is unlikely (high-income cities tend to have higher cinema ticket prices) or $\beta_3 = 0$, meaning $\log Y_i$ is irrelevant for explaining $\log T_i$ and was correctly omitted from the estimated regression (this case is trivial). Moreover, $\hat{\delta}_2$ will be inconsistent, even though $\widehat{\text{Cov}}\left(\log P_i, u_i\right) \xrightarrow{\text{p}} \text{Cov}\left(\log P_i, u_i\right) = 0$, since the limit of the second term above is not zero. Comparing the estimated regression (30) with an omitted variable and the (hypothetically) estimated DGP (31), we notice that the bias in $\hat{\delta}_1$ is positive (0.17 against the true value of -0.48), which is clear because $\beta_3 > 0$ and $\text{Cov}\left(\log P_i, \log Y_i\right) > 0$. Our original conjecture about the bias is confirmed.

(b) The researcher is not able to obtain data on Y_i but, from local authority records, she is able to obtain data on H_i, the average value of a house in each city. She fits the following regression:

$$\log T_i = \underset{(3.22)}{-0.63} - \underset{(0.22)}{0.37} \log P_i + \underset{(0.38)}{1.69} \log H_i, \quad R^2 = 0.36. \tag{34}$$

Describe the theoretical benefits from using H_i as a proxy for Y_i, discussing whether they appear to have been obtained in this example.

Solution: The proxy mitigates the problems caused by the omitted variable. Namely, the estimated coefficient on $\log P_i$ in (34) is very close to the true value - most of the positive bias that was present in (30) was corrected, but the coefficient appears to be insignificant. Also, $\log Y_i$ is a strong determinant of $\log T_i$. Compare the coefficient of determination R^2: 0.44 in regression (31) compared with 0.01 when $\log Y_i$ was omitted. Using $\log H_i$ as a proxy, the regression R^2 is 0.36, which is a substantial improvement. This suggests that $\log H_i$ is a reasonably good proxy for $\log Y_i$.

S5.7 Captain Obvious predicts: if you do well at school, you will do well at uni. A researcher investigating student performance at a US university estimates the following equation using a subsample of 3702 College students: these students were selected from the full sample if they had an SAT test score of 1200 or less.

$$\texttt{GPA}_i = \underset{(0.092)}{1.491} + \underset{(0.00008)}{0.001}\,\texttt{SAT}_i - \underset{(0.0006)}{0.013}\,\texttt{perc}_i + \underset{(0.019)}{0.157}\texttt{fem}_i - \underset{(0.040)}{0.314}\texttt{black}_i + \underset{(0.043)}{0.173}\texttt{athl}_i$$

$$R^2 = 0.2377, \qquad SSR = 1121.764$$

$\texttt{GPA}_i$ is the student's College Grade Point Average, $\texttt{SAT}_i$ is their score on a standardized College entry test, $\texttt{perc}_i$ is their percentile ranking in school (lower values indicate a better ranking), and $\texttt{fem}_i$, $\texttt{black}_i$ and $\texttt{athl}_i$ are dummy variables which indicate respectively whether the student concerned is female, black and engaged in student athletic activities.

The researcher is interested in testing whether his estimated equation can also forecast the performance of students with higher SAT scores (i.e. $\texttt{SAT}_i > 1200$). He starts

by selecting a high-scoring student at random: he adds data for this student to his sample, together with a dummy variable which is 0 for the original students, and 1 for the selected student, and re-estimates his original equation. The coefficient on the dummy variable is 0.521, with standard error 0.553.

(a) Explain how the researcher can use this result to conduct a formal test of the hypothesis that his original equation can forecast the outcome for this particular student, and conduct such a test using a 5% significance level.

Solution: Define a dummy variable

$$D_i = \begin{cases} 1 & \text{if } \texttt{SAT}_i > 1200 \\ 0 & \text{if } \texttt{SAT}_i \leq 1200 \end{cases}$$

Thus, for the new students, $i = 3703$, $D_i = 1$, and $D_i = 0$ for all $i = 1, \ldots, 3702$. Test for significance of the coefficient that corresponds to that dummy, $H_0 \colon \beta_D = 0$ against $H_1 \colon \beta_D \neq 1$ at a 5% level, where the OLS estimate is given by $\hat{\beta}_D = 0.521$ with standard error $s_{\hat{\beta}_D} = 0.553$. From OLS properties in large samples we know the sampling distribution of the following test statistic

$$T = \frac{\hat{\beta}_D - \beta_D^0}{S_{\hat{\beta}_D}} \overset{a}{\sim} N(0,1).$$

Reject H_0 for the two-sided test if the observed sample test statistic is such that $|t| > z_{0.975} = 1.96$. The observed sample test statistic is $t = 0.521/0.533 = 0.942 < 1.96$ and we do not reject the null at 5%. Conclude that there is insufficient evidence to suggest that this model cannot be used to forecast GPA for this particular student.

(b) The researcher now estimates two further equations, one for the 435 high-scoring students, and one for the combined sample of 4137 students. The regression output for the high-scoring students is given by

$$\widehat{\texttt{GPA}}_i = \underset{(0.578)}{0.914} + \underset{(0.0004)}{0.002}\,\texttt{SAT}_i - \underset{(0.002)}{0.020}\texttt{perc}_i + \underset{(0.060)}{0.127}\texttt{fem}_i - \underset{(0.329)}{0.294}\texttt{black}_i + \underset{(0.234)}{0.593}\texttt{athl}_i$$
$$R^2 = 0.2418, \qquad SSR = 129.911$$

The output for the regression for all students is

$$\widehat{\texttt{GPA}}_i = \underset{(0.078)}{1.306} + \underset{(0.00007)}{0.001}\,\texttt{SAT}_i - \underset{(0.0006)}{0.013}\,\texttt{perc}_i + \underset{(0.018)}{0.152}\texttt{fem}_i - \underset{(0.040)}{0.303}\texttt{black}_i + \underset{(0.043)}{0.205}\texttt{athl}_i$$
$$R^2 = 0.2978, \qquad SSR = 1259.999$$

Using this additional information, conduct a formal test (at 1% significance) of the hypothesis that the coefficients of the model are the same for both groups of students.

Solution: Conduct a Chow test for equality of the coefficients in the two regressions with $n_1 = 435$ (high-scoring students), $n_2 = 3702$ (low scoring students), and $n = 4137$ (common sample)

$$H_0 \colon \left(\beta_0^1 = \beta_0^2\right) \cap \cdots \cap \left(\beta_{\texttt{athl}}^1 = \beta_{\texttt{athl}}^2\right)$$
$$H_1 \colon \left(\beta_0^1 \neq \beta_0^2\right) \cup \cdots \cup \left(\beta_{\texttt{athl}}^1 \neq \beta_{\texttt{athl}}^2\right)$$

Use an F-test and significance level of 1%. The F-test statistic and its sampling distribution are given by

$$V = \frac{[SSR - (SSR_1 + SSR_2)] \,/\, (k+1)}{(SSR_1 + SSR_2) \,/\, (n - 2\,(k+1))} \sim F_{k+1,n-2(k-1)}$$

where SSR refers to the common sample, SSR_1 to regression for high-scoring students and SSR_2 to regression for low-scoring students, $k = 5$ is the number of regressors. The limiting distribution is $F_{6,4125}$, and so reject the null if $v > F_{0.99,6,4125} = 2.802$. The observed sample test statistic

$$v = \frac{[1259.99 - (1121.764 + 129.911)] \,/6}{(1121.764 + 129.911) \,/4125} = 4.567 > 2.802$$

Thus, H_0 is rejected at 1%. There is sufficient evidence to suggest the high/low SAT performers have different models (DGPs) for determining their GPA.